SpaceX Falcon 9 First Stage Landing Prediction

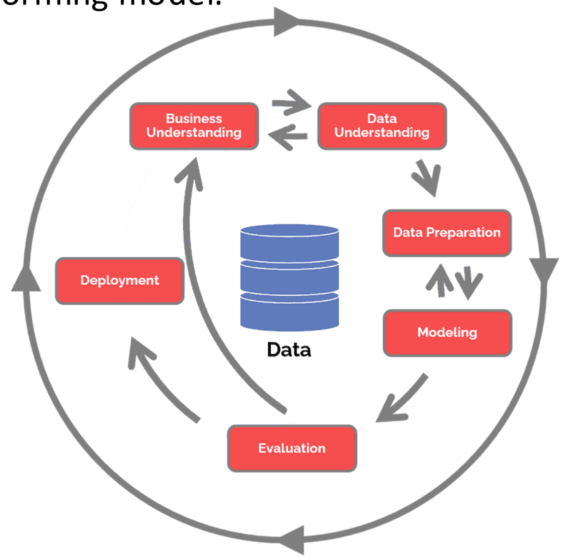
IBM Data Science Professional Certificate

Applied Data Science Capstone

Prepared by: Kulbhushan Ranaware

Executive Summary

- This project focuses on predicting the successful landing of the SpaceX Falcon 9 first stage using data science techniques.
- The project workflow includes data collection, data cleaning, exploratory data analysis, interactive visualization, and machine learning classification.
- Launch data was collected from the SpaceX API and public sources. Python, SQL, Folium, and Plotly Dash were used for analysis and visualization.
- Multiple machine learning models were developed to predict Falcon 9 first-stage landing success.
- The Support Vector Machine (SVM) achieved the highest prediction accuracy of 88%, making it the best-performing model.



Introduction

• SpaceX Falcon 9 is a reusable rocket system designed to reduce launch costs.

• Successful first-stage landing allows booster reuse and improves mission efficiency.

• Historical launch data can be analyzed to identify factors affecting landing success.

Project Objectives:

• Collect launch data
• Clean and prepare data
• Perform EDA
• Create visualizations
• Build prediction models

Data Collection Methodology

Data Sources:

• SpaceX REST API

• Public web sources

Tools Used:

• Python Requests

• Pandas

• Web Scraping

Collected Features:

• Launch site

• Payload mass

• Orbit type

• Booster version

• Landing outcome

Data Collection Evidence

The collected launch data was converted into a structured dataset using Python and Pandas.

The dataset contains important mission characteristics used for further analysis.

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
[9]
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.js
```
Python

We should see that the request was successfull with the 200 status response code

```
[10]
response=requests.get(static_json_url)
```
Python

```
[11]
response.status_code
```
Python

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
[12]
data = pd.json_normalize(response.json())
```
Python

Using the dataframe `data` print the first 5 rows

```
[13]
# Get the head of the dataframe
data.head()
```
Python

	static_fire_date_utc	static_fire_date_unix	tbd	net	window	rocket	success	details	crew	ships	capsules	
0	2006-03-17T00:00:00.000Z	1.142554e+09	False	False	0.0	5e9d0d95eda69955f709d1eb	False	Engine failure at 33 seconds and loss of vehicle	[]	[]	[]	[5eb0e4b5b6c3bb0C
1	NaN	NaN	False	False	0.0	5e9d0d95eda69955f709d1eb	False	Successful first stage burn and transition to second stage, maximum altitude 289 km, Premature engine shutdown at T+7 min 30 s, Failed to reach orbit, Failed to recover first stage	[]	[]	[]	[5eb0e4b6b6c3bb0C

Data Wrangling Methodology

Data preprocessing steps:

- Identified missing values
- Removed unnecessary columns
- Converted data types
- Processed categorical variables
- Prepared dataset for machine learning

Data Cleaning Results

The cleaned dataset improved data quality and ensured reliable analysis and model development.

Identify and calculate the percentage of the missing values in each attribute

```python
(df.isnull().sum()/len(df))*100
```

[3] ✓ 0.0s Python

```
FlightNumber      0.000000
Date              0.000000
BoosterVersion    0.000000
PayloadMass       0.000000
Orbit             0.000000
LaunchSite        0.000000
Outcome           0.000000
Flights           0.000000
GridFins          0.000000
Reused            0.000000
Legs              0.000000
LandingPad       28.888889
Block             0.000000
ReusedCount       0.000000
Serial            0.000000
Longitude         0.000000
Latitude          0.000000
dtype: float64
```

handling missing values

handling missing values

```python
df["LandingPad"] = df["LandingPad"].fillna("Unknown")
(df.isnull().sum()/len(df))*100
```

[4] ✓ 0.0s Python

```
FlightNumber      0.0
Date              0.0
BoosterVersion    0.0
PayloadMass       0.0
Orbit             0.0
LaunchSite        0.0
Outcome           0.0
Flights           0.0
GridFins          0.0
Reused            0.0
Legs              0.0
LandingPad        0.0
Block             0.0
ReusedCount       0.0
Serial            0.0
Longitude         0.0
Latitude          0.0
dtype: float64
```

EDA and Data Visualization Methodology

EDA was performed to identify relationships between launch characteristics and landing success.

Tools Used:

- Pandas
- Matplotlib
- Seaborn
- SQL

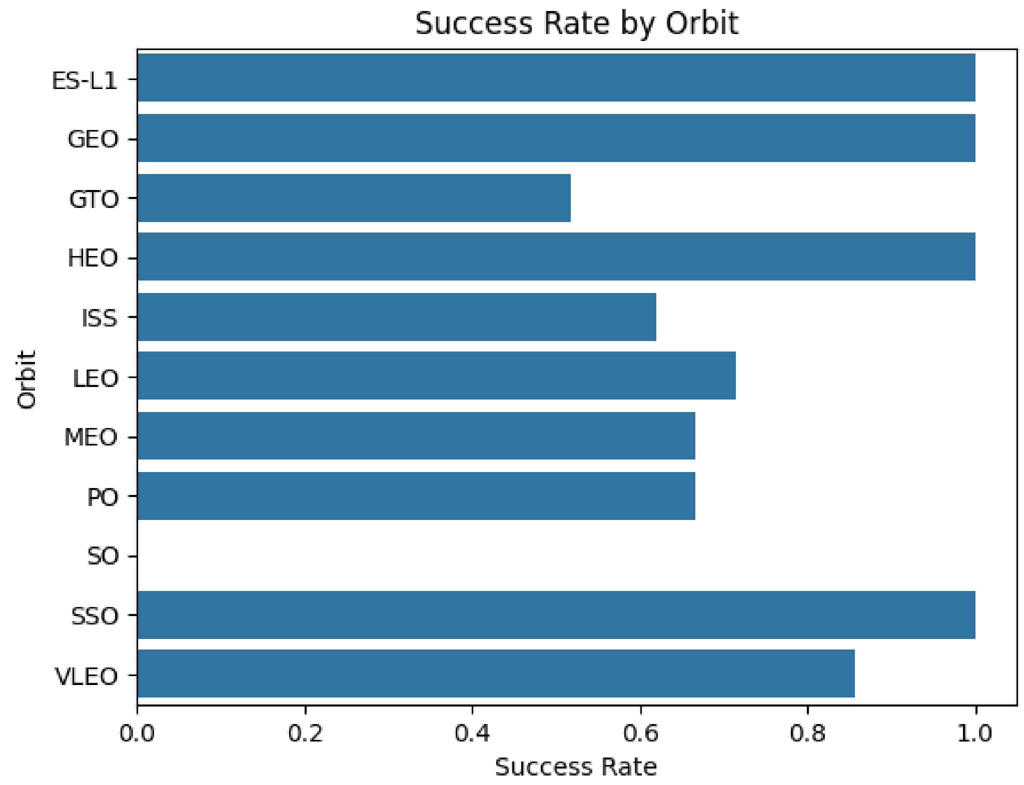
Analysis:

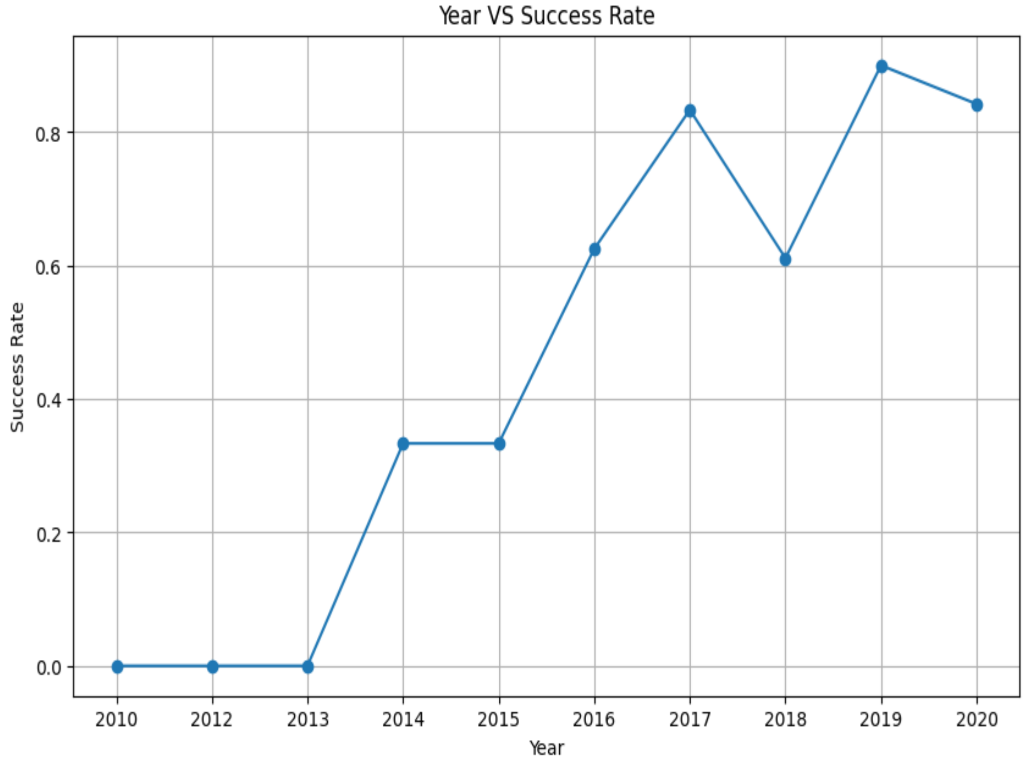
- Launch sites
- Payload mass
- Flight number
- Orbit type

Exploratory Data Analysis Results

Key observations:

- KSC LC-39A recorded the highest landing success rate.
- Landing success increased as booster flight experience increased.
- Payload mass affected the probability of a successful landing





EDA Using SQL Queries

SQL queries were used to analyze launch data.

Analysis included:

- Launch site performance
- Number of successful launches
- Payload information
- Mission characteristics

SQL Query Results

SQL analysis helped identify patterns in launch frequency and mission success.

Query 1: Total number of successful and failure missions.

```
Click to add a breakpoint
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Mission_Outcome;
""",conn)
```

Result: Mission outcomes were ranked by total count.

	Mission_Outcome	Total
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

Query 2: Rank landing outcomes by count from 2010-06-04 to 2017-03-20 in descending order.

```
Click to add a breakpoint
SELECT
    Landing_Outcome,
    COUNT(*) AS Count
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
""", conn)
```

Result: Landing outcomes were ranked by total count.

	Landing_Outcome	Count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

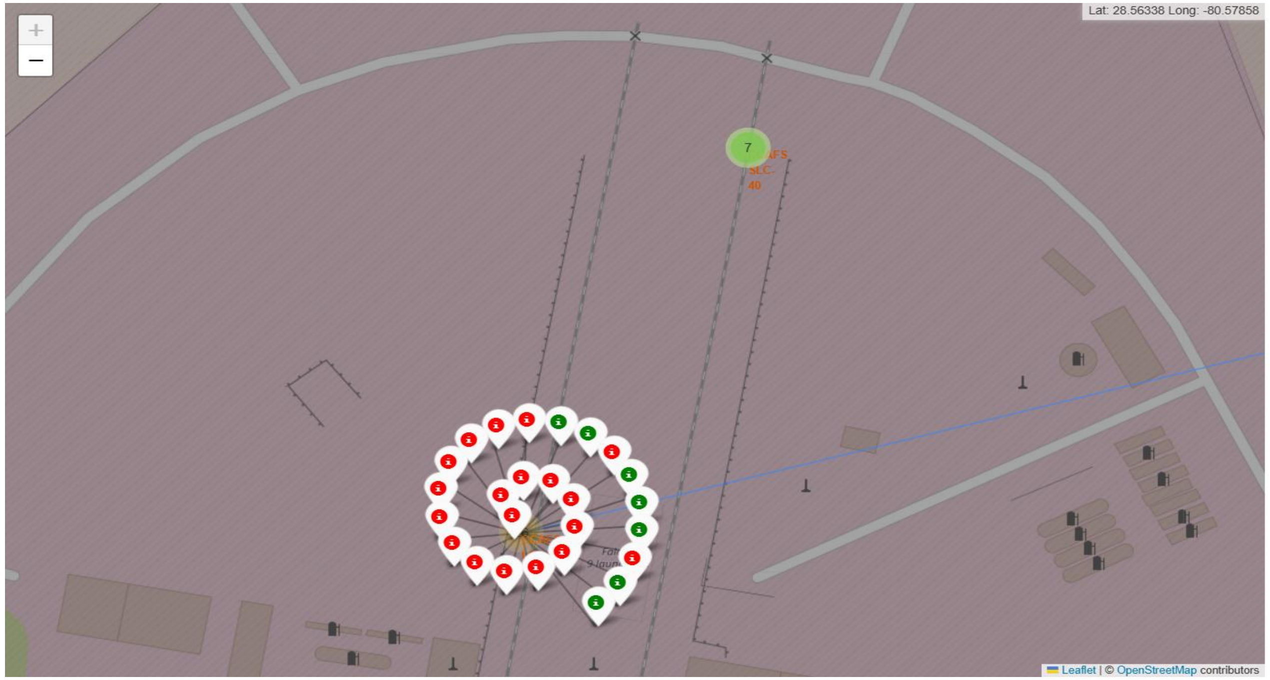
Launch Site Analysis Using Folium

Folium was used to create an interactive map of SpaceX launch locations.

The map helps analyze:

- Geographic distribution
- Launch site locations
- Success patterns

The interactive map shows that launch sites are concentrated along the U.S. east coast, enabling comparison of landing success by location.



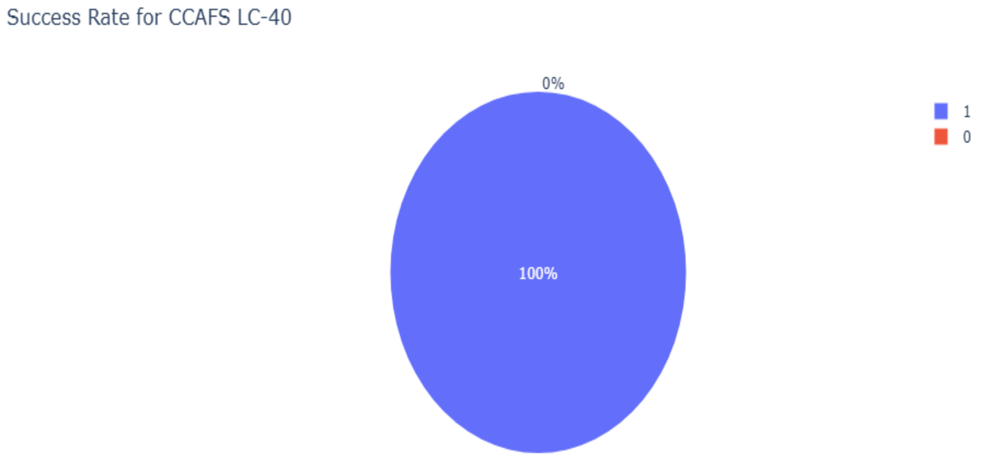
Interactive Dashboard Using Plotly Dash

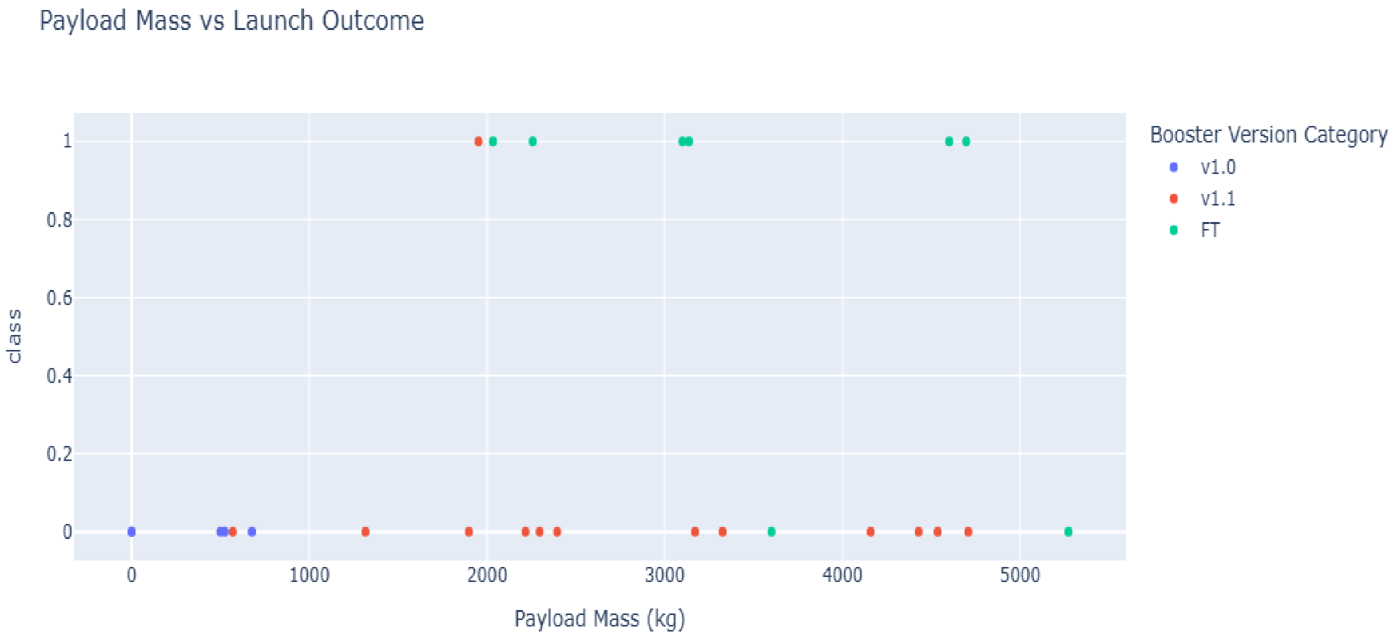
A Plotly Dash dashboard was developed for interactive exploration.

Features:

- Launch site filtering
- Payload selection
- Dynamic success visualization

The dashboard allows interactive filtering by launch site and payload mass, helping identify trends in landing success.





Predictive Analysis (Classification)

Machine learning models were developed to predict landing success.

Four classification algorithms were evaluated to determine the most accurate predictor of landing success.

Models:

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

Steps:

- Train-test split
- Feature scaling
- Model evaluation

Predictive Analysis Results

Model performance was compared using accuracy scores.

The best-performing model was selected based on prediction accuracy.

Model	Accuracy
Logistic Regression	0.83
Support Vector Machine (SVM)	0.88
Decision Tree	0.78
K-Nearest Neighbors (KNN)	0.83

SVM achieved the highest accuracy (88%) and was selected as the final model because it outperformed the other classifiers.

Conclusion

- Completed the complete data science workflow.
- Collected and cleaned SpaceX launch data.
- Performed EDA using Python and SQL.
- Created interactive visualizations.
- Developed machine learning models for landing prediction.
- This project demonstrates that machine learning can effectively predict Falcon 9 first-stage landing success using historical launch data.

Git hub

GitHub Repository

Repository:
https://github.com/kulbhushanranaware/IBM-Data-Science-Capstone

Repository includes:

- All completed Jupyter notebooks
- Plotly Dash notebook
- Folium notebook
- Machine Learning notebook
- Final Presentation

Thank You

Questions & Discussion